MAJOR PROJECT PHASE II · MID TERM EVALUATION

STRATEGIC APPLICATION OF ARTIFICIAL INTELLIGENCE IN CAD

SUBMITTED BY

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GUIDE

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Live demo
Scan, or open
aicad-vercel.vercel.app

WHERE PHASE I ENDED

A plain English requirement in, a sized, checked, modelled and drawn joint out

01 · AI
Read
Sentence to joint + load. Nothing invented is kept.

02
Size
Governing formula, rounded to a preferred size.

03
Check
Every failure mode vs allowable; iterate on failure.

04
Model
Exact solids, STEP files, assembly in position.

05
Draw
First angle A3 sheet, dimensions, title block.

6	44	3	290
JOINTS	FAILURE CHECKS	DESIGN SOURCES	AUTOMATED TESTS

FINAL ANALYSIS

Three errors found in published design literature, each traced, not assumed

FINDING	EVIDENCE
Knuckle eye shear: pin raised to 55 mm for bending, eye still checked with the original 40 mm pin	With the real pin: 80 MPa vs 65 allowed, 23% over its own limit
Cotter-spigot crushing never checked in the source	167 MPa vs 110 allowed at the source's own numbers; Leutwiler (1917) lists this mode independently, identical formula
Bolt tightening constant printed as 284*d* kN	Source's own M20 example only works out at 2840*d* N, a factor of ten out

None found by reading. All three surfaced because the system checks every failure mode on every run and prints the arithmetic, so a computed number disagreeing with a printed one was impossible to miss.

FINAL RESULTS OBTAINED

Cross-checked by an independent method, and against our own stated objectives

JOINT	PARTS	CHECKS	BUSIEST, % ALLOWABLE
Knuckle joint	5	9	76
Socket and spigot cotter joint	3	10	67
Sleeve and cotter joint	5	8	77
Gib and cotter joint	4	8	79
Hex bolt, nut and washer	3	2	71
Protected type rigid flange coupling	12	7	79

Finite element validation. 1.2% off a known case before trusting the method; 11.5% higher than the hand formula on the real eye rod, expected, because the formula ignores the stress concentration at the pin bore that the FEA sees directly.

5 of 6 stated objectives met without qualification. The sixth, opening the exported file inside NX10 itself, is verified in file structure (AP214 schema, 4 separate solids on the gib and cotter example), not yet watched opening on a lab machine.

FINAL ANALYSIS · THE CODE BEHIND THE CLAIM

The guard that stops the AI from inventing a number

```
def _traceable(value, text, factors=(1,)) -> bool:
    for n in _numbers_in(text):
        for f in factors:
            if abs(n * f - value) <= 0.01 * max(abs(value), 1e-9):
                return True
    return False

# used on every field the AI returns:
if not _traceable(v, text, factors):
    spec.notes.append(f"AI gave {key}={v:g}, which is not in the request; dropped")
    continue
```

Any value the model returns that cannot be matched back to a number actually present in the request, after unit conversion, is dropped and reported dropped. This is the literal mechanism behind "the AI only reads the sentence."

FINAL RESULTS OBTAINED · GOING DEEPER WITH AI

Where a part has no library entry, the AI writes the geometry itself, gated

```
# AI-written geometry for a part outside the 6-joint library, verified live:
# request: "a flat washer, 20 mm inner diameter, 40 mm outer diameter, 3 mm thick"
import cadquery as cq
result = cq.Workplane("XY").circle(20).circle(10).extrude(3).translate((0,0,-1.5))

# gate result, this run: 1 solid, valid, bbox (40.0, 40.0, 3.0) == declared (40, 40, 3)
```

Five gates before anything generated is trusted. Static screen (only cadquery/math, no file or network calls), separate process with a time limit, exactly one valid solid, bounding box matching what the AI declared, every size asked for present within 2 percent.

Re-run live on 16 Oct: attempt 1, PASS, one valid solid, bounding box (40.0, 40.0, 3.0) mm against a declared (40, 40, 3), volume 2827 mm³.

CONCLUSION AND FUTURE SCOPE

What exists today, what is proposed, kept clearly apart

Today. 6 joints, 44 checks, verified against three published sources, an independent finite element cross-check, 290 automated tests, a live web deployment.

Semester 8, proposed, not committed. Riveted and welded joints (same architecture), fatigue and stress concentration, tolerances and fits, the two unverified proportion sets, an NX journal script, a bill of materials.

Beyond the academic scope, as a direction, not a claim. The same translation layer, English requirement to verified design, applies to any catalogue of standard mechanical parts, which is what would make an industrial tie-up or a product around this pipeline realistic to propose once the joint library is broad enough.

PRESENTATION SKILLS AND VIVA-VOCE

Five sections, five presenters, each owning what they can defend

Ansh: the problem and the literature. Bhavya: what was built and the AI/engineering boundary. Atishay: the worked example, the output and the code. Dharmendra: the verification and the three findings. Ankur: validation, limits and the close.

On questions. Every claim traces to a file or a source and is said out loud if asked. Nothing is presented as more certain than it was checked to be.

PROJECT REPORT IN HARD COPY

Fifty pages in the department format, every claim traced to its source

50	8	12	6
PAGES	CHAPTERS PLUS APPENDIX	REFERENCES	OBJECTIVES, EACH ASSESSED IN SECTION 7.7

Built on the guide's own templates. Cover, certificate, declaration, acknowledgement, abstract, eight chapters, references and an appendix of screenshots; Section 7.7 scores each stated objective against evidence.

LIVE DEMO

Try it yourself: scan, or open aicad-vercel.vercel.app



Set the reader to "rules only" (or "AI, else rules"). The hosted copy has no AI key, so "AI only" will report an error.

Sentences that work: knuckle joint for 50 kN · gib and cotter joint for 50 kN · socket and spigot joint for 35 kN · flange coupling to transmit 20 kW at 1440 rpm.

Change the load and every dimension is recomputed, nothing is stored.